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6.2.4 Stability of a numerical algorithm

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ill-conditioned, then a small change in the input may result in a large change in the output regardless of how good your implementation is. Because you know you're going to incur error along the way. There might even have been error in the input when you measured it. So how do we quantify that we have a good implementation? Well, if we can show that for a value close to x let's call it x check, we actually get the exact same value as x

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Reading assignment

1/1 point (graded)

Read Unit 6.2.4 of the notes. [\[LINK\]](#)

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🗨️ How is the multiplication operation stable? For perturbed input, i would expect the product to be $(\chi + \delta \chi) * (\psi \dots$	4
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Homework 6.2.4.1

3/3 points (graded)

- Under the SCM from the last unit, floating point subtraction, $\kappa := \chi - \psi$, is a backward stable operation.

ALWAYS

✓ Answer: ALWAYS
- Under the SCM from the last unit, floating point multiplication, $\kappa := \chi \times \psi$, is a backward stable operation.

ALWAYS

✓ Answer: ALWAYS
- Under the SCM from the last unit, floating point division, $\kappa := \chi / \psi$, is a backward stable operation.

ALWAYS

✓ Answer: ALWAYS

Solution:

For complete solution, see the [notes](#).



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